

IMPORTANT NOTES

- Closed-book Exam : No phones on tables, PCs, nor Books.
- Bring your own stuff (Pen, Corrector, Pencil, Ruler, ...)
- Strictly, No talking to each other (Can I have a pen, ruler, charger, cable ... **NO**)
- Lecturers will **not answer nor discuss** any query during the exam.
- **Be brief** when answering the questions. Long paragraphs will be ignored.
- Bad or non-clear writing, will not be appreciated and you **will** be penalized.

Exercise 1 (General Questions : 5 points) :

For the following questions, a justification must be always provided, Yes/No answers will **not be** accepted unless a convincing explanation is given.

1. Is the language of all possible words composed over $\{a,b\}$ regular? True or False.
2. The epsilon transition (ϵ -transition) is allowed in Deterministic Finite Automata (DFA). True or False? .
3. This regular expression, $1^*(1^*01^*01^*)^*$, can be used to match binary strings that contain a number of zeros divisible by 3. True or False?
4. The language $L=\{wxzxw^R \mid w, x, z \in \{0,1\}^*, |w|>0, |x|>0 \text{ and } w^R \text{ is the reverse of } w\}$. L is **not** a regular language. True or False?
5. Is the language $L=\{a^n a^n \mid n \text{ is a natural number and } n \geq 0\}$ a regular language ? True or False?

Exercise 2 (Regular Languages : 6 points) :

1. [2 points] Provide the regular expressions for languages over the alphabet $\{0,1\}$:
 - a. Where all words in which each pair of 0s (i.e. **00**) is followed by a pair of 1s
 - b. Language whose words do not contain the substring **100** (! is not allowed)
2. [1 point] Using the pumping lemma, prove that the language $L=\{a^n b a^n \mid n \text{ is a natural number}\}$ is **not regular** over the alphabet $\{a,b\}$
3. [3 points] Draw strictly the deterministic finite automata (DFA) for the following languages :
 - a. $(001^+)^*$ (+ means 1 or more)
 - b. $L = \{ a^k b^m \mid k+m = 3n + 1 . k, m, n \in \{1, 2, 3, \dots\} \}$ over the alphabet $\{a,b\}$.

Exercise 3 (Context-Free Languages : 6 points) :

- [2 points] Create the context-free grammar (CFG) for the languages
 - $L = \{ a^i b^j c^k \text{ such that } j = i + k \} \text{ over } \{a,b,c\}$ (number of b=number of a+c)
 - $L = \{ w \# x \mid w^R \text{ is a substring of } x \text{ for } w, x \in \{a,b\}^* \}$
- [1 point] Create the PDA for $L = \{ w \in \{a, b\}^* : w \text{ has equal numbers of a's and b's} \}$.
- [1 point] Transform the following grammar into Chomsky Normal Form (CNF):
$$\begin{aligned} S &\rightarrow aSSA \mid ASC \mid a \\ A &\rightarrow AAB \mid B \mid C \\ B &\rightarrow a \mid \varepsilon \\ C &\rightarrow AC \mid CB \end{aligned}$$

- [2 points] Prove using the pumping lemma that $L = \{ 0^n 1^m \mid n \text{ and } m \text{ are natural numbers. } m \text{ is a multiple of } n \}$ is not a context-free language. 0^n means 0 being repeated n times.

Examples of words in the language L:

0011
0011**1111**
000111**111111**
000111**11111111**
00001111**1111**

Exercise 4 (Turing Machines : 3 points) :

Construct the turing machines for the following languages :

- [1 point] Construct a decider (NOT Transducer) turing machine to decide on whether two unary numbers separated by # are sorted such that the small number is written first (on the left side of the #). The alphabet used is only {1}.
For instance :
111#1 should be **rejected**. The first number has more digits/characters.
1#1111 should be **accepted**. The first number has less digits/characters.
11#11 should be **accepted**. Both numbers have the same number of characters.

Don't bother with cases where a number is not given (111#), the two numbers are strictly given to you. No need to swap.

- [2 points] Write the algorithm (1 point) and construct the decider Turing machine (1 point) that decides on the language :

$$A = \{ 0^{2^n} \mid n \geq 0 \}$$

The language consists of all strings of 0s whose length is a power of 2.

Examples :

00 (n=1, 2 zeros)
0000 (n=2, 4 zeros)
00000000 (n=3, 8 zeros)
0000000000000000 (n=4, 16 zeros)